

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist and materials scientist with 6+ years of experience synthesizing, processing, and characterizing functional material systems, using spectroscopy, thermal analysis, electrochemistry, CVD/surface preparation, and DOE to connect process conditions with structure-property-performance. Currently leading polymer process development at CU Boulder while bringing prior thin-film coatings, interface, and data-driven experimental design experience from Georgia Tech and ETH Zurich.

CORE COMPETENCIES

Materials Synthesis & Process Development

Functional Materials Characterization

Thin Films, Coatings & Interfaces

Spectroscopy & Thermal Analysis

CVD & Surface Preparation

Structure-Property-Processing Relationships

DOE & Experimental Design

Data-Driven Experimental Execution

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead process-development experiments for bio-based polymer systems, using DOE to define synthesis/process windows, improve reproducibility, and translate material-performance targets into executable experimental plans.
- Build multi-technique characterization workflows using FT-IR, thermal analysis, and spectroscopy to connect processing decisions with material structure, quality, and performance outcomes.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and synthesized functional organic electrochromic material systems, then established structure-property relationships through FT-IR, UV-Vis-NIR, electrochemical, TGA, and DSC characterization.
- Investigated degradation, adhesion, and interface-instability mechanisms in thin-film systems, converting failure-analysis results into material design and process recommendations.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded development of molecular thin-film coatings for optical hardware, designing surface-preparation and CVD deposition strategies to tune morphology, adhesion, and device-relevant performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed and optimized polymer thin-film deposition processes using DOE and in-situ QCM-D monitoring, establishing process controls that improved coating reproducibility and uniformity.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed a 15-member research laboratory operating multiple analytical instruments and parallel materials projects, implemented equipment tracking and calibration systems, and improved instrument uptime by 30%.
- Authored SOPs and coordinated technical workflows across concurrent projects with tight quality and timeline requirements.

Georgia Institute of Technology

2019 - 2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates and developed supplemental teaching materials that improved exam performance by 15%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Functional Materials Synthesis & Characterization Chemistry of Materials 2025

Synthesized and characterized electrochromic materials, linking molecular design and processing decisions to optical, electrochemical, thermal, and stability performance.

Organic synthesis, FT-IR, UV-Vis-NIR, electrochemistry, TGA, DSC

AFOSR Defense Thin-Film Coatings DoD Funded

Developed molecular thin-film coatings for optical hardware, tuning CVD and surface-preparation conditions to achieve target morphology, adhesion, and performance.

CVD, surface chemistry, coatings, materials characterization

Bio-Based Polymer Process Development Current

Lead DOE-guided process-development experiments for bio-based polymer systems, improving reproducibility, material quality, and translation from target properties to experimental plans.

DOE, process optimization, polymer chemistry, spectroscopy

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials Synthesis & Characterization: Functional organic materials, polymer systems, thin films, coatings, CVD, surface preparation, FT-IR, UV-Vis-NIR, ellipsometry, TGA, DSC, QCM-D, NMR, spectroelectrochemistry

Structure-Property Analysis: Structure-property-processing relationships, degradation analysis, adhesion/interface troubleshooting, root-cause analysis, data interpretation

Process & Data: DOE, SPC, statistical data analysis, process optimization, technical documentation, experimental planning

Execution & Communication: Multi-project management, cross-functional communication, stakeholder updates, scientific writing

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)